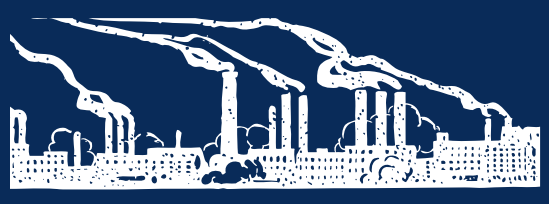
OCEAN ACIDIFICATION

CHANGING OCEAN CHEMISTRY ON HAWAIIAN SHORES

OCEAN ACIDITY IS GREATER THAN IT HAS EVER BEEN IN THE PAST 2 MILLION YEARS[1]

Burning fossil fuels releases carbon dioxide into the atmosphere[2]



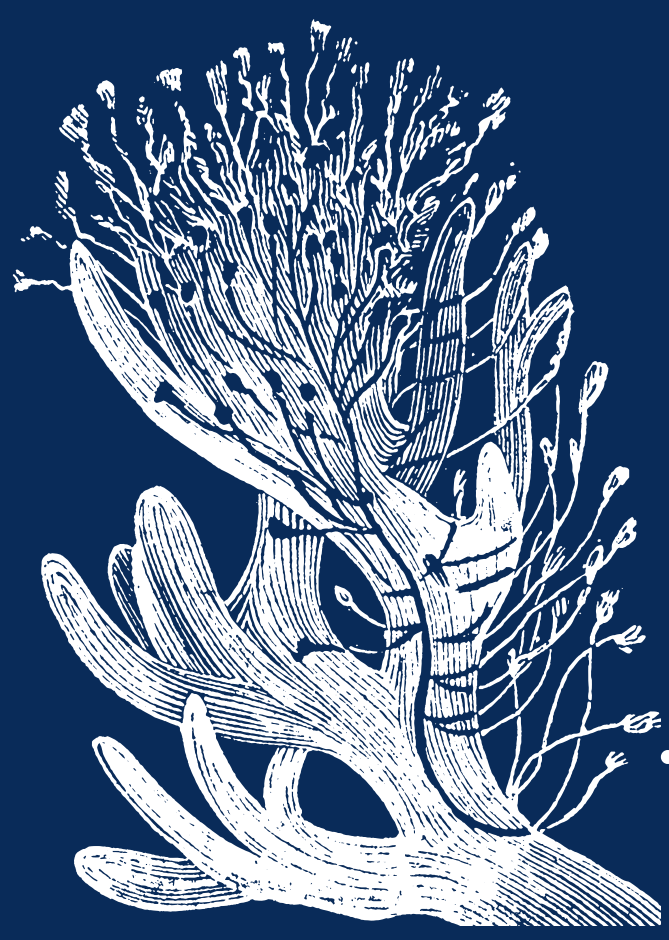
26% of CO_2/year

from human activities is absorbed by the ocean, making oceans more acidic[3]

Alters marine ecosystems and disrupts sea life[2]

BY THE NEXT CENTURY...

ocean acidity is expected to be 1.5x higher than pre-industrialization levels[4]



WHAT ARE THE IMPACTS ON OCEAN LIFE?

- **10-25% reduction in skeletal densities of shelled-sea animals**[5]
- **Reduces ocean biodiversity and ecosystem resilience**[5]

HOW WILL THIS AFFECT HAWAI'IS COMMUNITY?

- **Estimated $10.6 billion USD loss from the tourism industry this century**[6]
- **Threatens sea creatures that have intrinsic value in Hawaiian culture**[7]

SOLUTIONS

1. **Growing native corals & plants (which can absorb CO_2) in restoration sites & nature reserves & transplanting them into the ocean**[8]
2. **Integrating Hawaiian values regarding respect for the land into tourist activities, by hiring more native Hawaiians with traditional knowledge**[9]

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